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Master 1 – Intelligent Computer Systems Engineering (SII)

Module: Introduction to Image Processing (ITI)

A Comparative Study of RAG Techniques for Generating Image Captions

Phase-1 Technical Report: Data Exploration & Pre-processing

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Comprehensive empirical report integrating teacher clarifications: 10,000 KB + 100 Val from `train2017` (10,100 total), 1,000 Test from `val2017` with 100 comparison subset, image-grounded annotations, and Step 5 confidence policy.

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1 Executive Summary & Project Step-by-Step Structure

Automatic image caption generation requires generating fluent, semantically accurate descriptions for input visual scenes. Standard Vision-Language Models (VLMs) frequently hallucinate non-existent entities, omit fine-grained peripheral objects, or confuse spatial relationships. Retrieval-Augmented Generation (RAG) resolves these deficiencies by grounding caption generation in external, verified knowledge evidence (semantically similar visual examples and human-validated reference captions).

This report documents the rigorous completion of **Phase 1: Data Exploration and Pre-processing**. To ensure total scientific transparency and reproducibility, the project pipeline is organized into ten distinct, sequentially executed steps:

Table 1: Step-by-Step Methodological Architecture of Phase 1

Step	Pipeline Phase	Core Objective & Implemented Operation	Section
Step 1	Data Acquisition	Download and verify official MS COCO 2017 raw archives and models.	§2
Step 2	Split Partitioning	Create 10,000 KB and 100 Val from `train2017`; 1,000 Test from `val2017`.	§3
Step 3	Data Auditing	Full Pillow decoding, SHA-256 integrity, duplicate checks, anomaly detection.	§4
Step 4	Visual Pre-processing	Bicubic resizing, central 224×224 crop, RGB normalization for CLIP.	§5
Step 5	Caption Pre-processing	Whitespace normalization (`clean`) while preserving `raw` text.	§5
Step 6	Metadata Extraction	Visual bounding box grounding; human confidence = 1.0 convention.	§6
Step 7	Statistical EDA	Distribution of objects, scene complexity, vocabulary, and spatial relations.	§7
Step 8	Multimodal Embeddings	CLIP ViT-B/32 dense 512-dimensional L2-normalized vector extraction.	§8
Step 9	FAISS Vector Indexing	Exact cosine similarity search via `IndexFlatIP` ($O(N)$ dot product).	§8
Step 10	Validation Retrieval	Zero-leakage retrieval verification on 100 validation queries; qualitative case study.	§9

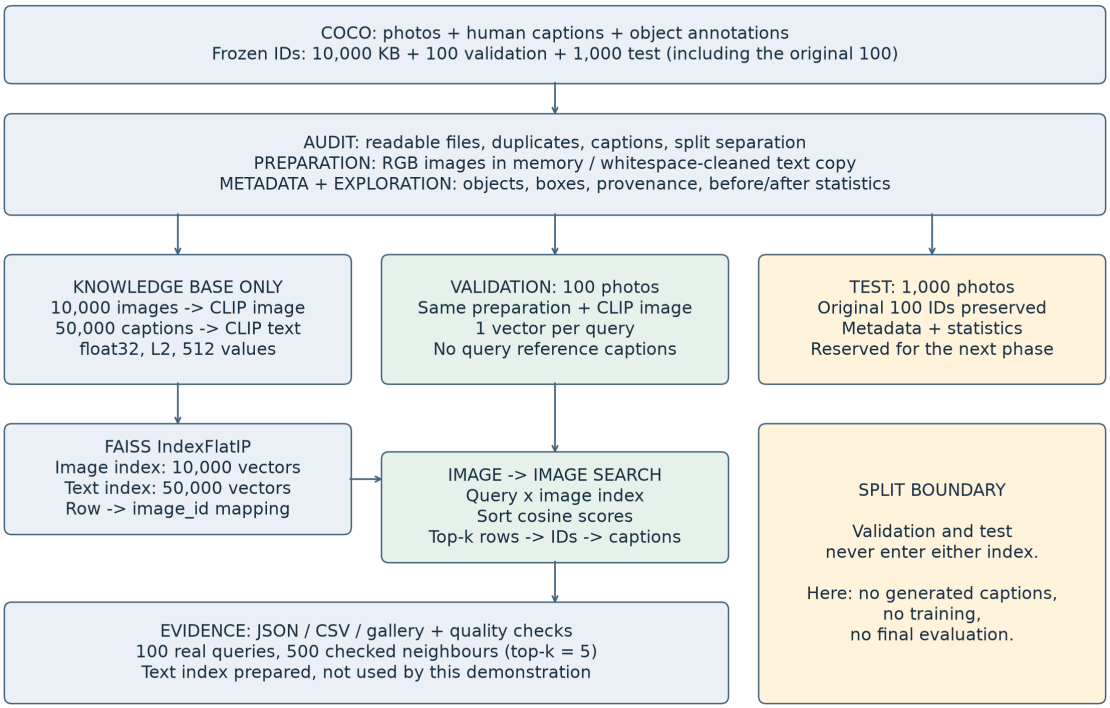


Figure 1: End-to-end Phase 1 execution architecture. Arrows denote data flow; test reference captions remain strictly isolated from retrieval.

2 Step 1: Environment Setup & Data Acquisition

The project relies on a modular, reproducible software environment built on Python 3.12. All dependencies are frozen and version-pinned in `requirements.txt` (including `torch 2.14.1+cpu`, `transformers 5.21.0`, `faiss-cpu 1.15.1`, `nltk 3.10.4`, and `pycocotools 2.0.11`).

2.1 Raw Data Acquisition

Raw data are acquired directly from official Microsoft COCO S3 cloud endpoints via HTTPS. The download step retrieves:

1. `annotations_trainval2017.zip`: containing human captions (`captions_train2017.json`, `captions_val2017.json`) and object bounding boxes (`instances_train2017.json`, `instances_val2017.`
2. **Image Archives**: Images corresponding to the frozen splits are downloaded and verified into `data/raw/train2017/` and `data/raw/val2017/`.
3. **Pretrained Models**: The vision-language encoder `openai/clip-vit-base-patch32` is downloaded and pinned to immutable commit revision `3d74acf9a28c67741b2f4f2ea7635f0aaf6f0268`.
4. **NLP Resources**: The English grammatical POS tagger (`averaged_perceptron_tagger_eng`) is cached locally in `data/interim/cache/nltk_data/`.

3 Step 2: Dataset Partitioning & Anti-Leakage Protocol

In strict compliance with the instructor's clarifications, the dataset partitions are constructed to guarantee zero information leakage between training, tuning, and evaluation:

Table 2: Exact Dataset Partition Allocation and Sources

Split Partition	COCO Origin	Target	Audited	Captions
Knowledge Base (KB)	`train2017`	10,000	10,000	50,000
Validation Subset	`train2017`	100	100	500
Preserved Comparison Test	`val2017`	100	100	500
Extended Evaluation Test	`val2017`	1,000	1,000	5,000
Total Distinct `train2017`	`train2017`	**10,100**	**10,100**	**50,500**
Total Overall Dataset	`train + val`	**11,100**	**11,100**	**55,500**

3.1 Mathematical Formulation of Mutual Exclusion

Let $\mathcal{U}_{\text{train}}$ represent the eligible image pool from `train2017`, and $\mathcal{U}_{\text{val}}$ represent the eligible pool from `val2017`. Partitioning satisfies:

$$\mathcal{S}_{\text{val}} \subset \mathcal{U}_{\text{train}}, \quad |\mathcal{S}_{\text{val}}| = 100 \tag{1}$$

$$\mathcal{S}_{\text{KB}} \subset (\mathcal{U}_{\text{train}} \setminus \mathcal{S}_{\text{val}}), \quad |\mathcal{S}_{\text{KB}}| = 10,000 \implies \mathcal{S}_{\text{KB}} \cap \mathcal{S}_{\text{val}} = \emptyset \tag{2}$$

The total number of images extracted from `train2017` is exactly $10,000 + 100 = \mathbf{10,100}$.

3.2 Test Set Preservation and Comparison Protocol

Initial course materials referenced a test set of 100 images, while the teacher clarified that 1,000 test images from `val2017` should be prepared, keeping the original 100 test images for comparative analysis. To fulfill both requirements without bias:

1. The original 100 test images are frozen in `artifacts/splits/comparison/test_100_ids.json`.
2. The 1,000-image test set (`artifacts/splits/test_ids.json`) was created by retaining the 100 original images and augmenting them with 900 additional stratified images from $\mathcal{U}_{\text{val}} \setminus \mathcal{S}_{\text{test-100}}$:
$$\mathcal{S}_{\text{test-100}} \subset \mathcal{S}_{\text{test-1000}} \subset \mathcal{U}_{\text{val}}, \quad |\mathcal{S}_{\text{test-1000}}| = 1,000 \tag{3}$$
3. Stratified sampling is performed across three scene complexity strata: Simple (0–3 objects), Medium (4–7 objects), and Complex ($\geq$ 8 objects) using seed 42 and largest-remainder proportional allocation.

4 Step 3: Data Auditing, Integrity Checks & Anomaly Handling

Before running any preprocessing or statistical aggregation, an exhaustive data audit was executed across all 11,100 image files and 55,500 captions.

Table 3: Data Quality Control Audit Results and Integrity Assertions

Audit Check Parameter	Observed Count	Verification Status
Expected image files across splits	11,100	Target
Images present on local disk	11,100	100% Verified
Images successfully opened and decoded by Pillow	11,100	100% Valid
Missing, unreadable, or truncated image files	0	Fully Passed
Captions per image	Exactly 5	5.00 average (zero variance)
Shared image IDs between splits ($\mathcal{S}_{\text{KB}} \cap \mathcal{S}_{\text{val}}$)	0	Anti-Leakage Verified
Shared image IDs with test ($\mathcal{S}_{\text{KB}} \cap \mathcal{S}_{\text{test}}$)	0	Anti-Leakage Verified
Identical file or pixel hash duplicates	0	Zero Duplication
Identified degenerate bounding boxes	1	Documented & Handled
Identical normalized caption strings across splits	87	Inspected Non-Leaking

4.1 Documented Anomaly Resolutions

1. **Degenerate Bounding Box (Image ID 200365)**: In COCO image `200365`, annotation ID `918` specifies bounding box $[296.65, 388.33, 1.03, 0.0]$, exhibiting a height of $h = 0.0$. Modifying or deleting official COCO annotation files would compromise benchmark comparability. The system retains the image, captions, and category label, but marks the bounding box with `bbox_valid = false`. This degenerate box is excluded from spatial geometry and IoU calculations.

2. **Cross-Split Identical Caption Text**: Exactly 87 normalized caption strings appear identically across separate images in different splits (e.g., generic descriptions such as *"A group of people sitting at a dining room table."* or *"A train traveling down train tracks at night."*). Audit confirmed that these represent natural linguistic overlap between independent real-world photographs, with zero image ID or annotation ID leakage.

5 Steps 4 & 5: Visual and Caption Pre-processing

5.1 Step 4: Image Pre-processing for Vision-Language Encoders

Image pre-processing formats the raw variable-resolution COCO JPEGs into standard input tensors for the CLIP encoder:

1. **Color Mode & EXIF Transposition**: Images are loaded and converted to 3-channel RGB. EXIF orientation tags are applied via `ImageOps.exif_transpose` to prevent inverted or rotated tensor inputs.

2. **Bicubic Aspect-Preserving Resizing**: The shorter dimension is scaled to 224 pixels.

3. **Central Cropping**: A centered 224×224 crop is extracted.

4. **Channel Standardization**: Intensities are scaled to $[0, 1]$ and normalized by CLIP's channel statistics:

$$\mu = [0.48145466, 0.45782750, 0.40821073], \quad \sigma = [0.26862954, 0.26130258, 0.27577711] \quad (4)$$

   Producing tensors of shape $(1, 3, 224, 224)$ with finite, non-NaN floating-point values.

5.2 Step 5: Caption Normalization (Before vs. After Analysis)

To prevent evaluation leakage, raw reference captions are preserved uncorrupted for test metrics (BLEU, CIDEr, SPICE), while normalized versions are prepared for retrieval:

- `caption_raw`: Preserves original punctuation, casing, and spacing.
- `caption_clean`: Normalizes repeated spaces, leading/trailing whitespace, and line breaks.

Table 4: Before vs. After Text Normalization Statistics Across Partitions

Split Partition	Total Captions	Modified	Raw Characters	Clean Characters
Knowledge Base	50,000	7,743	2,620,933	2,612,839
Validation	500	72	25,725	25,645
Test (1,000)	5,000	745	262,492	261,674
Test Subset (100)	500	88	25,680	25,584

6 Step 6: Metadata Extraction & Ground-Truth Confidence Policy

Note on Section Numbering: This phase implements Step 5 ("Metadata extraction") from Section 4.4.3 of the instructor's project brief. It is numbered as Step 6 in our technical report to reflect the exact sequential 10-step implementation pipeline.

Extracted metadata are structured as line-delimited JSON (`JSONL`) files (`artifacts/metadata/{split}.jsonl`). Each record maps one image to its file properties, SHA-256 fingerprint, five captions, visual objects, bounding boxes, derived spatial relations, and attribute heuristics.

6.1 Confidence Policy for Step 5 (Teacher Specification): The Ground-Truth Convention

In strict accordance with the instructor's project guidelines for Step 5:

- **Human Reference Confidence**: Because COCO reference captions are human-generated ground-truth descriptions, they are assigned:

$$\texttt{source} = \text{"coco_human_reference"}, \quad \texttt{confidence} = 1.0, \quad \texttt{threshold} = 1.0 \tag{5}$$

  This is documented in `config.yaml` and `metadata.py` as a *provenance convention*. It denotes that human ground-truth data is accepted with 100% confidence, rather than representing a model detection probability.

- **Image Grounding vs. Captions**: Objects are extracted directly from COCO instance annotations with image-space bounding box coordinates (`source: "coco_annotation"`), rather than extracted purely as nouns from caption text.

- **Heuristic Attributes**: Lexicon-matched attributes (e.g., color keywords such as *"white"* or *"blue"*) carry `source = "caption_heuristic"` and do not inherit confidence = 1.0.

7 Step 7: Statistical Exploratory Data Analysis (EDA)

Per the required methodological flow, **statistics start directly from the extracted objects** (*Extraction* → *Exploration*). All figures and tables are computed from the structured metadata records.

7.1 Object Class Distribution & Class Imbalance

The 10,000-image Knowledge Base contains 72,507 individual object annotations across all 80 COCO categories. Table 5 details the 25 most frequent categories:

Table 5: Top 25 Most Frequent Object Classes in the Knowledge Base (72,507 total objects)

Rank	Category	Instances	Rank	Category	Instances	Rank	Category	Instances
1	**person**	22,670	10	**handbag**	1,044	18	**sheep**	829
2	**car**	3,815	11	**bird**	950	19	**truck**	751
3	**chair**	3,151	12	**banana**	945	20	**potted plant**	752
4	**bottle**	2,086	13	**umbrella**	920	21	**motorcycle**	721
5	**book**	1,986	14	**bench**	869	22	**backpack**	706
6	**cup**	1,584	15	**boat**	829	23	**cow**	704
7	**dining table**	1,288	16	**wine glass**	673	24	**knife**	687
8	**bowl**	1,158	17	**donut**	662	25	**kite**	620
9	**traffic light**	1,112	–	–	–	–	–	–

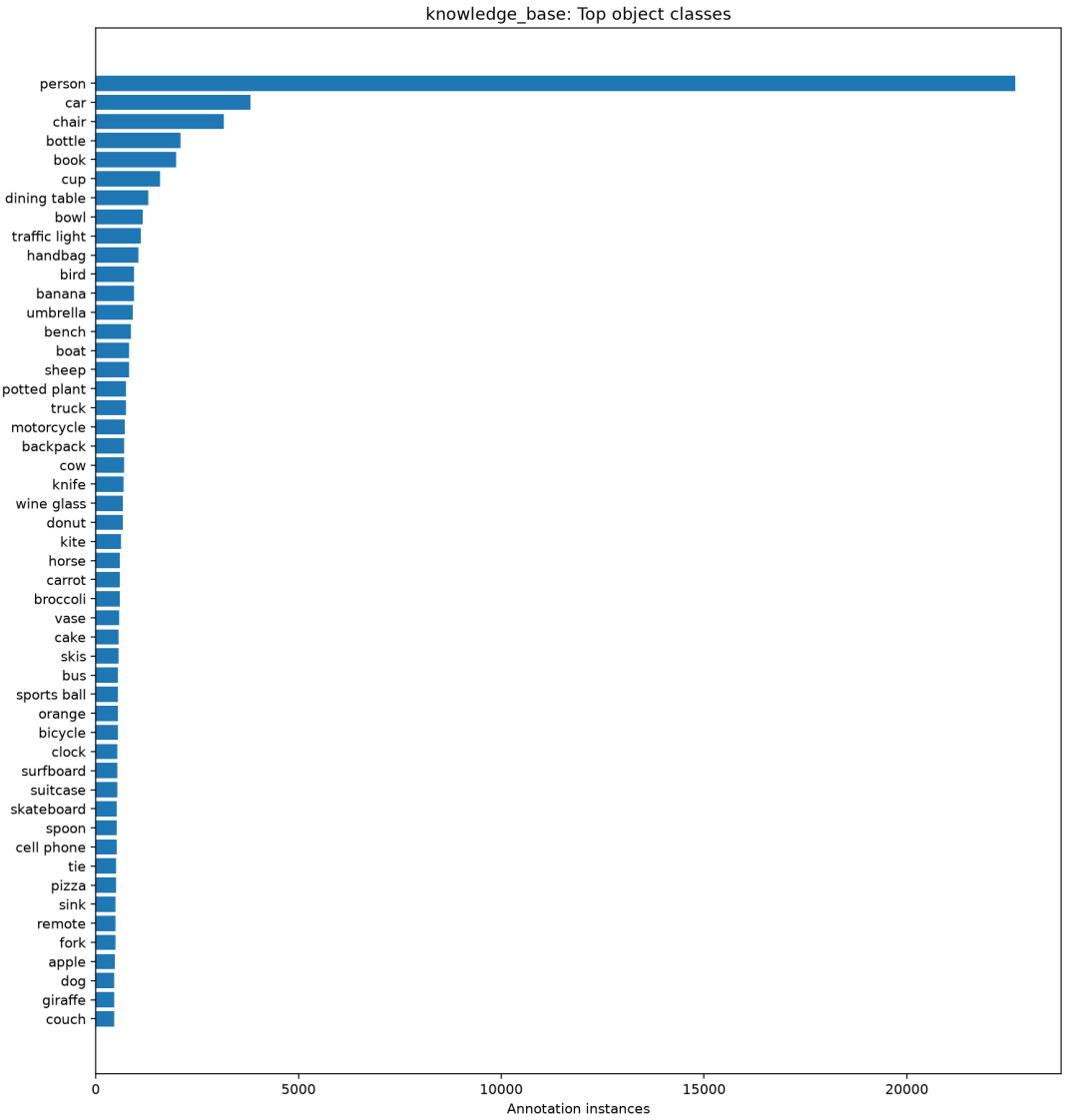


Figure 2: Frequency distribution of the top 50 object categories in the 10,000-image Knowledge Base.

The distribution demonstrates significant class imbalance: `person` accounts for 31.3% of all objects, while rare categories like `toaster` (17) and `hair drier` (20) appear infrequently.

7.2 Scene Complexity and Partition Comparative Analysis

Scene complexity is quantified by the number of visual object instances per image:

Table 6: Comprehensive Scene Complexity Comparison Across All Partitions

Complexity Metric	**Knowledge Base**	**Validation**	**Test (100)**	**Test (1,000)**
Total Images	10,000	100	100	1,000
Mean Objects / Image	7.2507	7.3400	7.0500	7.3880
Median Objects / Image	4.0	4.5	4.0	4.0
Population Standard Deviation	7.3305	8.3848	6.4628	7.4213
Minimum – Maximum Range	0 – 59	0 – 58	1 – 26	0 – 63
Simple Scenes (0–3 objects)	4,247 (42.47%)	41 (41.00%)	42 (42.00%)	422 (42.20%)
Medium Scenes (4–7 objects)	2,449 (24.49%)	25 (25.00%)	24 (24.00%)	238 (23.80%)
Complex Scenes ($\geq$ 8 objects)	3,304 (33.04%)	34 (34.00%)	34 (34.00%)	340 (34.00%)

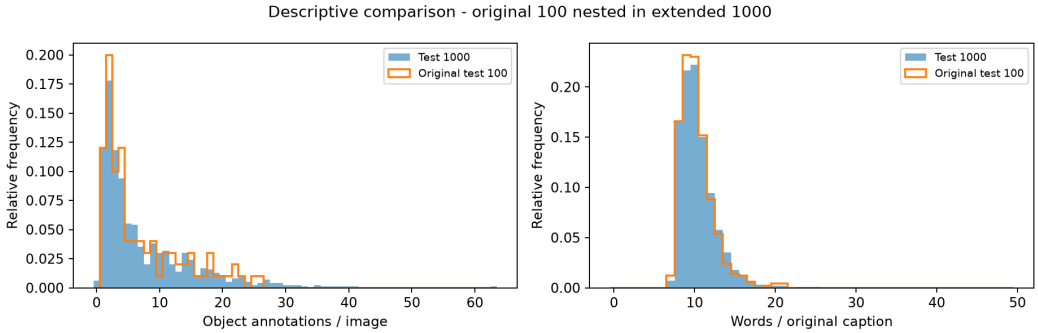


Figure 3: Normalised statistical comparison between Test-100 and the extended Test-1000 set.

The close alignment between Test-100 and Test-1000 distributions validates the proportional stratification algorithm, confirming that the extended test set preserves complexity proportions without skewing distribution parameters.

7.3 Textual Statistics and POS Linguistic Analysis

Caption length and vocabulary distributions were analyzed using NLTK's Treebank tokenizer:

Table 7: Textual Statistics and Lexical Diversity Across Partitions

Linguistic Property	**Knowledge Base**	**Validation**	**Test (100)**	**Test (1,000)**
Reference Captions	50,000	500	500	5,000
Mean Words / Caption	10.4525	10.2900	10.2720	10.4490
Median Words / Caption	10.0	10.0	10.0	10.0
Caption Length Range	6 – 49	7 – 32	7 – 21	7 – 49
Unique Vocabulary Tokens	9,972	945	904	3,372
Total Word Tokens	522,625	5,145	5,136	52,245

Table 8: Top Part-of-Speech (POS) Tags in the Knowledge Base (Penn Treebank Tagset)

Tag	Description	Token Count	%	Tag	Description	Token Count	%
NN	Singular noun	149,690	28.6%	**VBN**	Verb past participle	6,693	1.3%
DT	Determiner	107,020	20.5%	**CD**	Cardinal number	6,334	1.2%
IN	Preposition/conjunction	83,586	16.0%	**TO**	Infinitival "to"	5,743	1.1%
JJ	Adjective	41,620	8.0%	**VBP**	Verb non-3rd person	5,524	1.1%
NNS	Plural noun	30,721	5.9%	**RP**	Adverbial particle	3,671	0.7%
VBG	Verb gerund/participle	29,990	5.7%	**PRP$**	Possessive pronoun	3,646	0.7%
VBZ	Verb 3rd person singular	13,688	2.6%	**RB**	Adverb	3,525	0.7%
CC	Coordinating conjunction	11,967	2.3%	**PRP**	Personal pronoun	3,356	0.6%
NNP	Proper noun singular	6,916	1.3%	**WDT**	Wh-determiner	2,308	0.4%

7.4 Object Co-occurrence & Spatial Relationships

- **Spatial Relations**: Evaluated across all ordered pairs of valid bounding boxes. The Knowledge Base contains 495,256 ordered `left_of` / `right_of` pairs, 495,260 `above` / `below` pairs, and 45,020 box overlap pairs (IoU ≥ 0.1).
- **Derived Attribute Keywords**: The most frequent color mentions in captions are *"white"* (1,997 images), *"black"* (1,123), *"red"* (870), and *"blue"* (780). Common spatial prepositions include *"on"* (7,283 images), *"near"* (1,361), and *"behind"* (532). Context keywords include *"outside"* (563 images), *"inside"* (400), and *"night"* (136).

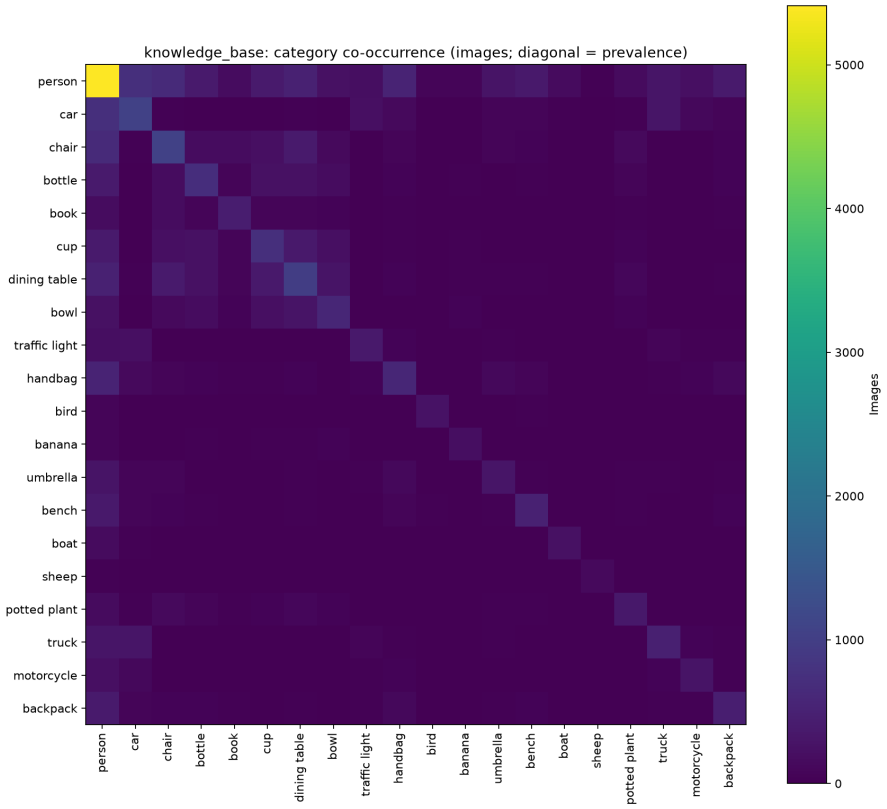


Figure 4: Co-occurrence heatmap among the top 20 object categories.

8 Steps 8 & 9: Multimodal Embedding Generation & FAISS Indexing

8.1 Step 8: CLIP Dense Representation

Embeddings are generated using the pretrained `openai/clip-vit-base-patch32` encoder:

- **Image Vectors**: 10,000 images are processed in batches of 32 on CPU, yielding a tensor of shape $(10000, 512)$ saved to `artifacts/embeddings/image.npy`.
- **Text Vectors**: 50,000 captions are tokenized and encoded, yielding a tensor of shape $(50000, 512)$ saved to `artifacts/embeddings/text.npy`.
- **L2 Normalization**: Every vector is normalized to unit length:

$$v = \frac{\mathbf{e}}{\|\mathbf{e}\|_2} \implies \|v\|_2 = 1.000000 \tag{6}$$

8.2 Step 9: FAISS Vector Database Construction

FAISS is configured with `IndexFlatIP` (exact inner product search). Because all stored and query vectors are L2-normalized, the inner product is mathematically identical to Cosine Similarity:

$$\langle q, v \rangle = \sum_{j=1}^{512} q_j \cdot v_j \equiv \frac{q \cdot v}{\|q\|_2 \|v\|_2} = \cos(\theta_{q,v}) \tag{7}$$

Table 9: FAISS Index Specifications and Performance Metrics

Modal Modality	Index Algorithm	Indexed Vectors	Dimensions	Build Latency	Disk Foot
Image Index	`IndexFlatIP`	10,000	512	0.0075 s	20.4
Caption Index	`IndexFlatIP`	50,000	512	0.0340 s	102.4

Row mappings (`image_rows.json` and `text_rows.json`) preserve direct, verified mappings between vector matrix rows, COCO image IDs, and reference captions.

9 Step 10: Validation Retrieval & Qualitative Error Analysis

To validate end-to-end retrieval, all 100 disjoint validation images were encoded as visual queries ($k = 5$ neighbors retrieved) against the 10,000-image knowledge base index. Validation reference captions were strictly withheld during search. Average query retrieval latency was 4.167 ms on CPU across the 500 retrieved neighbors.

9.1 In-Depth Case Study: Validation Image 490

Validation image `000000000490.jpg` depicts a dog standing with its front paws on a skateboard inside a dimly lit room (indoor scene with hardwood/parquet floor, wall, and dark shadows).

Table 10: Top-5 Nearest Neighbors Retrieved for Validation Query 490

Rank	Retrieved COCO ID	Cosine Similarity	Primary Retrieved Reference Caption
1	267664	0.818381	*"A lone cat sits in a dark room with some furniture."*
2	545323	0.801236	*"A dog is sitting on a black bench."*
3	127643	0.790362	*"A brown dog standing on top of a skateboard."*
4	516396	0.778416	*"A dog standing with front paws on a skateboard"*
5	81686	0.772747	*"A dog sits on a rug with its eyes closed."*



Figure 5: Qualitative retrieval results for validation query 490.

9.2 Qualitative Analysis: Why RAG Is Needed

- **Visual Success**: Ranks 3 and 4 successfully retrieved the exact compound semantics: *"dog + skateboard"* (*"A brown dog standing on top of a skateboard"* at cosine similarity 0.790362, and *"A dog standing with front paws on a skateboard"* at 0.778416). Rank 2 also correctly identified a dog (*"A dog is sitting on a black bench"*).
- **Failure Mode at Rank 1**: Rank 1 retrieved an image containing a *cat* in a dark room rather than a dog. Global CLIP vision embeddings pool scene-level features (ambient lighting, dark shadows, parquet floor reflections, and animal silhouette). Because query 490 is set in a dimly lit indoor room, CLIP's global visual similarity score assigned the highest score (0.818381) to a dark room containing a cat and furniture.
- **Phase 2 Roadmap**: This failure mode illustrates why RAG with reranking, multimodal query expansion, or metadata filtering is necessary. In Phase 2, visual object verification will filter out candidates referencing a cat when visual analysis detects a dog.

10 Pipeline Reproduction Commands & Phase 2 Roadmap

10.1 Terminal Reproduction Protocol

The pipeline is reproducible via modular CLI commands:

Listing 1: Complete Step-by-Step CLI Reproduction Commands

```
# 1. Environment installation
python -m venv imp
./imp/Scripts/python.exe -m pip install -r requirements.txt

# 2. Step 1: Download annotations, images, NLP models, and CLIP encoder
python -m src.cli download-annotations
python -m src.cli download-images
python -m src.cli download-nlp
python -m src.cli download-model

# 3. Steps 2, 3 & 6: Data audit and metadata extraction
python -m src.cli audit
python -m src.cli metadata

# 4. Step 7: Statistical Exploratory Data Analysis (EDA)
python -m src.cli eda

# 5. Steps 8 & 9: Multimodal embeddings and FAISS index construction
python -m src.cli --offline embeddings
python -m src.cli index

# 6. Step 10: Validation retrieval and quality control
python -m src.cli --offline validate-retrieval --top-k 5
python -m src.cli --offline qc
python -m src.cli report
```

10.2 Phase 2 Transition Roadmap

With Phase 1 data, metadata, and FAISS indices frozen and validated, the foundation is complete for Phase 2:

1. **Generative VLM / LLM Integration**: Prompting generative language models with retrieved multimodal context.
2. **Assigned RAG Technique Implementation**: Applying the assigned technique (e.g., Naive RAG, Reranking, Multimodal RAG, Self-RAG, or Metadata-Filtered RAG).
3. **Quantitative Evaluation**: Evaluating generated captions against COCO reference captions using standard metrics: BLEU-4, METEOR, ROUGE-L, CIDEr, and SPICE.
4. **Statistical Hypothesis Testing**: Performing Shapiro-Wilk normality tests, Student's paired t-tests or Wilcoxon signed-rank tests with Bonferroni correction to confirm statistically significant improvements.